Problem 4

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$$F_n = \begin{cases} 1 & \text{if } n = 1 \\ 1 & \text{if } n = 2 \\ F_{n-1} + F_{n-2} & \text{if } n \geq 3 \end{cases}$$

Proposition 1.

$$\forall m \in \mathbb{N}, \sum_{j=1}^{m} F_{2j-1} = F_{2m}$$

Proof. The proof is by induction on m.

BASE CASE: Consider the case for $m = 1$. Observe,

$$\begin{aligned} \sum_{j=1}^{1} F_{2j-1} &= F_{2(1)-1} \\ &= F_{2-1} \\ &= F_1 \\ &= 1 \end{aligned}$$

and,

$$F_{2(1)} = F_2 = 1$$

Thus, the result holds for the base case $m = 1$.

INDUCTIVE CASE: Suppose the result holds for some natural number k. That is,

$$\sum_{j=1}^{k} F_{2j-1} = F_{2k}$$

Now consider the case for $k + 1$. Observe,

$$\begin{aligned}\sum_{j=1}^{k+1} F_{2j-1} &= \sum_{j=1}^{k}(F_{2j-1}) + F_{2(k+1)-1} \\ &= F_{2k} + F_{2(k+1)-1} \\ &= F_{2k} + F_{2k+1} \\ &= F_{2k+2} \\ &= F_{2(k+1)}\end{aligned}$$

Let's pull back the curtain on some of the magic here. $\sum_{j=1}^{k}(F_{2j-1}) = F_{2k}$ by the inductive hypothesis. And $F_{2k} + F_{2k+1} = F_{2k+2}$ by the definition of the Fibonacci sequence. We are allowed to do this substitution because the index $2k+2$ is always greater than 3, meaning that F_{2k+2} sits within the $n > 3$ piece of F_n.

The result, $F_{2(k+1)}$ has the form F_{2m} for $m = k+1$. Thus, the result holds in the inductive case. Therefore, by the principle of mathematical induction, the result holds for all $m \in \mathbb{N}$.

□

Proposition 2.

$$\sum_{j=1}^{m} F_{2j} = F_{2m+1} - 1$$

Proof. The proof is by induction on m.

An impressive proof.

□